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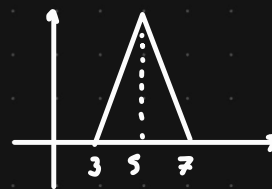
EQUATIONS:

• TRIANGULAR $T_r(x)$



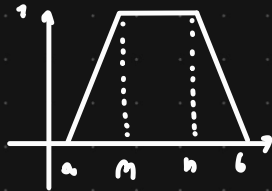
$$\begin{cases} \frac{x-a}{m-a} & \text{if } a < x \leq m \\ \frac{b-x}{b-m} & \text{if } m < x \leq b \\ 0 & \text{OTHERWISE} \end{cases}$$

SAME SIDE

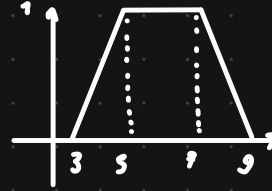


$$\begin{cases} \frac{x-3}{5-3} & \text{if } 3 < x \leq 5 \\ \frac{7-x}{7-5} & \text{if } 5 < x \leq 7 \\ 0 & \text{OTHERWISE} \end{cases}$$

• TRAPEZOIDAL



$$\begin{cases} \frac{x-a}{m-a} & \text{if } a < x \leq m \\ 1 & \text{if } m < x \leq n \\ \frac{b-x}{b-m} & \text{if } n < x \leq b \\ 0 & \text{OTHERWISE} \end{cases}$$



$$\begin{cases} \frac{x-3}{5-3} & \text{if } 3 < x \leq 5 \\ 1 & \text{if } 5 < x \leq 7 \\ \frac{9-x}{9-7} & \text{if } 7 < x \leq 9 \\ 0 & \text{OTHERWISE} \end{cases}$$

• NORMAL: $\sup(f(x)) = 1$?

• SUPPORT: $\{x \in X \mid f(x) > 0\}$

• CORE: $\{x \in X \mid f(x) = 1\}$

• α -CUT: $\{x \in X \mid f(x) \geq \alpha\}$

if $\alpha = 1 \Rightarrow$ CORE

• STRONG α -CUT: $\{x \in X \mid f(x) > \alpha\}$

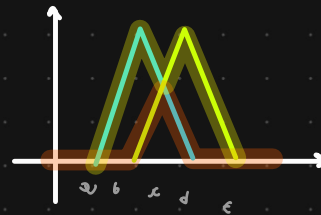
• CARDINALITY: $|A|$ $\leftarrow$ ONLY FOR DISCRETE

• ENTROPY H: $H = - \sum_{i=1}^n p_i \log_2(p_i) + (1-p_i) \log_2(1-p_i)$

• MEASURE OF FUZZYNESS:

• w/ HAMMING DISTANCE: $1 - \frac{(|p_1 - p_1^*| + |p_2 - p_2^*| + \dots + |p_n - p_n^*|)}{\| \text{SUPP}(y) \|}$

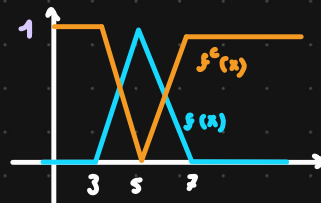
• w/ EUCLIDEAN DIST: $1 - \frac{\sqrt{(p_1 - p_1^*)^2 + (p_2 - p_2^*)^2 + \dots + (p_n - p_n^*)^2}}{\| \text{SUPP}(y) \|}$



• INTERSECTION (MIN) $\leftarrow$ T-NORM

• UNION (MAX) $\leftarrow$ T-CONORM

$$f^c(x) = 1 - f(x)$$



$$f(x) = \begin{cases} \frac{x-3}{5-3} & \text{if } 3 < x \leq 5 \\ \frac{7-x}{7-5} & \text{if } 5 < x \leq 7 \\ 0 & \text{OTHERWISE} \end{cases}$$

$$f^c(x) = \begin{cases} \frac{5-x}{5-3} & \text{if } 3 < x \leq 5 \\ \frac{x-7}{7-5} & \text{if } 5 < x \leq 7 \\ 1 & \text{OTHERWISE} \end{cases}$$

ENTROPY BASE 2

• COMPLEMENT of g :

$$\begin{aligned} g(10) &= 0.3 & g^c(10) &= 0.7 & j(10) &= 0.7 \\ g(15) &= 0.5 & g^c(15) &= 0.5 & j(15) &= 0.4 \\ g(20) &= 1 & g^c(20) &= 0 & j(20) &= 0 \\ g(25) &= 1 & g^c(25) &= 0 & j(25) &= 1 \\ g(30) &= 0.5 & g^c(30) &= 0.5 & j(30) &= 1 \\ g(35) &= 0.3 & g^c(35) &= 0.7 & j(35) &= 0.3 \\ g(x) &= 0 & g^c(x) &= 1 & j(x) &= 0 \end{aligned}$$

• SUPPORT of g : $\{10, 15, 20, 25, 30, 35\}$

• CARDINALITY of g $CARD(g) = 0.3 + 0.5 + 1 + 1 + 0.5 + 0.3 = 3.6$

• α -CUT with $\alpha > 0.7$ $\{15, 20, 25, 30\}$

• ENTROPY of g $-\sum_{x \in X} g(x) \log(g(x)) + (1-g(x)) \log(1-g(x))$
 $-(0.3 \log(0.3) + 0.5 \log(0.5) + 1 \log(1) + 1 \log(1) + 0.5 \log(0.5) + 0.3 \log(0.3) + 0.7 \log(0.7) + 0.7 \log(0.7) + 0.7 \log(0.7) + 0.7 \log(0.7) + 0.7 \log(0.7) + 0.7 \log(0.7))$

COMPLEMENTARY $\rightarrow$

• FUZZYNESS WITH HAMMING DISTANCE

$$1 - \frac{|0.3 - 0.7| + |0.5 - 0.5| + |1 - 0| + |1 - 0| + |0.5 - 0.5| + |0.3 - 0.7|}{||SUPP(g)||}$$

• FUZZYNESS WITH EUCLIDEAN DISTANCE

$$1 - \frac{\sqrt{(0.3 - 0.7)^2 + (0.5 - 0.5)^2 + (1 - 0)^2 + (1 - 0)^2 + (0.5 - 0.5)^2 + (0.3 - 0.7)^2}}{||SUPP(g)||}$$

• ORDERED WEIGHTED AVERAGE $w_i = 1/6$

$$(1 \cdot \frac{1}{6}) + (1 \cdot \frac{1}{6}) + (0.5 \cdot \frac{1}{6}) + (0.5 \cdot \frac{1}{6}) + (0.3 \cdot \frac{1}{6}) + (0.3 \cdot \frac{1}{6})$$

• CORE of $\{20, 25\}$

• VERY g : $g^2(x)$
 • ENOUGH g : $\sqrt{g(x)}$
 • SURELY g : $2g^2(x)$

• CENTER OF GRAVITY

$$\frac{10 \cdot 0.3 + 15 \cdot 0.5 + 20 \cdot 1 + 25 \cdot 1 + 30 \cdot 0.5 + 35 \cdot 0.3}{3.6 \leftarrow CARD(g)}$$

• OWA $w_1 = 0.75$
 $w_2 = 0.25$

$$\begin{aligned} \bullet OWA(g, g)(10) &= 0.7 \cdot 0.75 + 0.3 \cdot 0.25 \\ \bullet OWA(g, g)(15) &= 0.5 \cdot 0.75 + 0.5 \cdot 0.25 \\ \bullet OWA(g, g)(20) &= 1 \cdot 0.75 + 0 \cdot 0.25 \\ \bullet OWA(g, g)(25) &= 1 \cdot 0.75 + 0 \cdot 0.25 \\ \bullet OWA(g, g)(30) &= 1 \cdot 0.75 + 0.5 \cdot 0.25 \\ \bullet OWA(g, g)(35) &= 0.3 \cdot 0.75 + 0.3 \cdot 0.25 \\ \bullet OWA(g, g)(x) &= 0 \text{ OTHERWISE} \end{aligned}$$

$$v(p) = 0.7 \quad v(q) = 0.4$$

TRUTH VALUE

AND

$$\begin{aligned} \bullet MIN: MIN(p, q) &= 0.4 \\ \bullet LUKASIEWICZ: MAX(q, p+q-1) &= 0.1 \\ \bullet PRODUCT T-NORM: p \cdot q &= 0.7 \cdot 0.4 = 0.28 \end{aligned}$$

OR

$$\begin{aligned} \bullet MAX: MAX(p, q) &= 0.7 \\ \bullet LUKASIEWICZ: MIN(p+q, 1) &= 1 \\ \bullet PRODUCT T-COORN: p+q-p \cdot q &= 0.7 + 0.4 - 0.28 = 0.82 \end{aligned}$$

$p \rightarrow q$

$$\bullet CODEL: \begin{cases} q & \text{if } p > q \\ 1 & \text{OTHERWISE} \end{cases} \rightarrow q = 0.4 \iff MIN(1, q)$$

$$\bullet GOUVEN: \begin{cases} q/p & \text{if } p > q \\ 1 & \text{OTHERWISE} \end{cases} \rightarrow q/p = 0.57$$

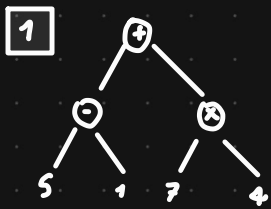
$$\bullet LUKASIEWICZ: \begin{cases} 1 & \text{if } p \leq q \\ 1-p+q & \text{OTHERWISE} \end{cases} \rightarrow 1-p+q = 0.7 \iff MIN(1-p+q)$$

$\neg p$

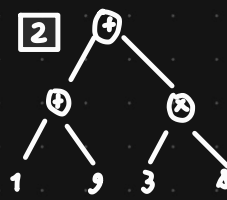
$$\begin{aligned} \bullet LUKASIEWICZ: 1-p &= 0.3 \\ \bullet CODEL: NEG(p) &= 0 \\ \bullet GOUVEN: NEG(p) &= 0 \end{aligned}$$

$\{ \begin{matrix} 1 & p \leq q \\ 0 & p > q \end{matrix} \}$

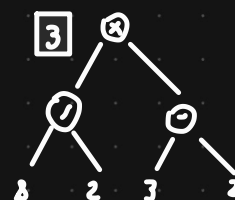
$$f(x) = x + 2$$



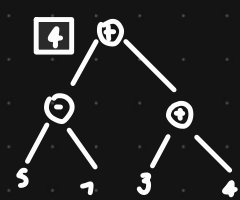
$$f(x) = 34$$



$$f(x) = 36$$

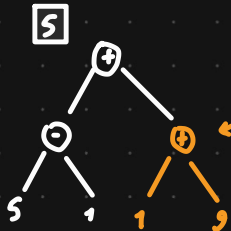


$$f(x) = 6$$



$$f(x) = 13$$

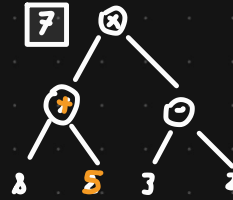
1 AND 2 ARE MOST FIT



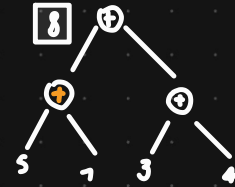
$$f(x) = 16$$



$$f(x) = 54$$



$$f(x) = 15$$



$$f(x) = 13$$

SELECTION: REMOVE 1 2 3 AND 4

$$f(x) = x_3 + 2x_2 + 6x_1 + x_0$$

IND. 1	0100	$f(1) = 2$	$P(\text{IND. 1}) = 2/20 = 0.1$	PICK IND. 1 <u>0</u> TIMES
IND. 2	1011	$f(2) = 8$	$P(\text{IND. 2}) = 8/20 = 0.4$	PICK IND. 2 <u>2</u> TIMES
IND. 3	1010	$f(3) = 7$	$P(\text{IND. 3}) = 7/20 = 0.35$	PICK IND. 3 <u>2</u> TIMES
IND. 4	0101	$f(4) = 3$	$P(\text{IND. 4}) = 3/20 = 0.15$	PICK IND. 4 <u>0</u> TIMES

$$f(\text{TOT}) = 20$$



MUTATION



AGE BASE SELECTION
KEEP IND. 5, IND. 6, IND. 7
IND. 8 AND REMOVE IND. 1,
IND. 2, IND. 3 AND IND. 4

SELECTION

ROULETTE WHEEL SELECTION

- FITNESS(A) = 3 $3/6 = 0.5$ CHOOSE A 2 TIMES
 - FITNESS(B) = 1 $1/6 = 0.17$ CHOOSE B 0 TIMES
 - FITNESS(C) = 2 $2/6 = 0.33$ CHOOSE C 1 TIME
- TOT. FITNESS: 6

FITTEST SELECTION $f(x) = x^2$ SELECT THE FITTEST INDs

TOURNAMENT SELECTION $K = 2$

CHOOSE K INDIVIDUALS WITH HIGHEST FITNESS

MUTATION

MUTATION RATE CHANGE A BIT INDEPENDENTLY FROM THE OTHERS WITH PROB P_M

0100 $P_M = 1/4 \rightarrow 0101$

SWAP MUTATION PICK TWO ALLELES AT RANDOM AND SWAP THEM (MUTATIONS)

123456789 $\rightarrow$ 153426789

INSERT MUTATION

123456789 $\rightarrow$ 125346789

SCRAMBLE MUTATION PICK A SUBSET AND REARRANGE RANDOMLY

123456789 $\rightarrow$ 153246789

INVERSION MUTATION PICK A SUBSET AND INVERT IT

123456789 $\rightarrow$ 154326789

AGE BASE SELECTION REMOVE THE OLDEST GENERATION
GENERATIONAL REPLACEMENT

PERMUTATIONS

CROSSOVER 1-POINT

123|45 $\longrightarrow$ 12321

543|21 $\longrightarrow$ 54345

ORDER 1 CROSSOVER FOLLOW THE ORDER OF THE
OTHER INDIVIDUAL (FOR ADDING THE MISSING NUMBERS)

123456789 $\longrightarrow$ 382456719

937826514 $\longrightarrow$ 347826591

PARTIALLY MAPPED CROSSOVER

NO

123456789 $\longrightarrow$ 4567

937826514 $\longrightarrow$ 3265

CYCLE CROSSOVER IDENTIFY CYCLE

123456789 $\longrightarrow$ 137426589

937826514 $\longrightarrow$ 923856714

EDGE RECOMBINATION

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